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RESEARCH ARTICLE

Deep and Domain Specific Feature-Based Cervical Cancer Classification Using Support Vector Machine Optimized With Particle Swarm Optimization

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ABSTRACT Cervical cancer presents a significant threat to the global healthcare system and its early detection is a challenging task. Cervical cancer is detected through manual evaluation of Pap smear images by expert pathologists. Computer-aided diagnosis (CAD) systems developed using machine learning and deep learning-based algorithms present a promising solution. This work proposes a novel approach by combining the deep features extracted from MobileNetV1 model with the domain-specific shape features, texture features and color features for cervical cancer classification. The noisy features have been eliminated by utilising the concept of mutual information whereas, a support vector machine optimized with Particle Swarm Optimization (PSO) was used for the classification. The proposed methodology is evaluated on the publicly available Sipakmed dataset, comprising 4049 sample images across five different classes. The suggested approach obtains an accuracy of 97.9% in Pap Smear cell image categorisation task. Thus, it presents a promising approach for the cervical cell categorisation task.

INDEX TERMS MobileNetv1, cervical cancer classification, Hu moments, Haralick features, Zernike moment, mutual information, feature fusion.

I. INTRODUCTION

Cervical cancer stands as a significant global health issue, being ranked as fourth most widespread cancer affecting women. World Health Organization reports that an estimated 660,000 new cases are diagnosed annually, with a staggering mortality rate of approximately 350,000 deaths [1]. Despite the concerning burden it presents, early detection of cervical cancer significantly increases the chances of successful prevention and treatment [2]. Over 80% of new cases and a staggering 85% of deaths attributed to cervical cancer occur due to restricted access to screening initiatives and treatment services [3].

A screening methodology of cervical cancer is Pap Smear that involves microscopic analysis of cell samples collected from the cervix [4]. While effective, it has inherent

limitations. Firstly, the manual evaluation process is laborious and time-consuming. Cytotechnologists must meticulously examine slides containing upwards of 100,000 cells stained using the Papanicolaou method [5]. Secondly, this visual assessment is susceptible to subjective interpretation, leading to potential inter-observer variability and misdiagnosis [6]. These limitations, coupled with the high cost associated with manual screening, can hinder accessibility of population-wide cervical cancer prevention programs. The limitations of traditional Pap smear analysis, including subjectivity and inter-observer variability, have spurred research into alternative methods. ML and DL techniques have surfaced as promising tools for cervical cancer screening by automating image analysis and potentially improving diagnostic accuracy.

Various studies have investigated different ML approaches to automate the cervical cancer classification from images. Sukumar et al. [7] suggested a computer based diagnostic

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system utilizing a hybrid classifier and extracting six sets of morphological features for classification with an SVM. Focusing on interpretability, Kumar et al. [8] developed a computational methodology for automated recognition and labeling utilizing biologically interpretable features, incorporating K-means clustering for segmentation, and subsequently employing K-nearest neighbor (KNN) classification. Another approach [9] employed a two-level cascade system integrating C4.5 and logistic regression (LR) classifiers for automatic cancer detection. ML algorithms like Random Forest (RF) have also been explored. Another study [10] developed a classification model using RF by leveraging known risk factors associated with cervical cancer, combined with SMOTE (Synthetic Minority Oversampling Technique) to address class imbalance and PCA (Principal Component Analysis) is utilized for dimensionality reduction. However, ML-based approaches have exhibited limitations, including lower accuracy in classifying specific cell types, dependence on manual feature engineering, which can be tedious, prone to individual bias, and error prone [11].

Among DL algorithms, Convolutional Neural Networks (CNNs) have proved their capability in image classification tasks, including applications in medical image analysis. CNNs leverage three key concepts: local receptive fields, weight sharing, and pooling. A plethora of studies have explored the use of CNNs for cervical cancer classification. Wu et al. [12] suggested a DL model with a relatively simple architecture for classifying cervical cancer from cytological images. This model has the potential to assist pathologists in improving diagnostic accuracy, particularly in regions with limited access to healthcare resources. Ghoneim et al. [13] investigated a DL solution that combines a CNN for feature extraction with an Extreme Learning Machine (ELM) classifier. This approach utilizes strengths of CNNs in learning image features and the efficiency of ELMs for classification. Harangi et al. [14] investigated an approach that incorporates segmentation predictions from Fully Convolutional Networks (FCNs) alongside the original image data. This approach utilizes FCNs to generate refined segmentation maps that can improve classification accuracy. Building upon pre-trained models like AlexNet and VGG-16, Kudva et al. [15] introduced a novel transfer learning technique. Their method involves constructing a CNN and initializing its weights based only on filters identified as relevant for cervical cancer detection from the pre-trained networks. Despite the promising results achieved with CNNs, challenges remain. Traditional end-to-end DL models often require significant computational resources and infrastructure, limiting their applicability in settings with limited technological capabilities.

This paper introduces HFF-Net, a Hybrid Feature Fusion Network designed for accurate and efficient cervical cell classification using Pap smear images. The key contributions of this work are:

- 1) Hybrid Feature Representation: HFF-Net utilizes a pre-trained MobileNetV1 for extraction of high-level image

features. Complementing the abstract deep features, HFF-Net incorporates morphological and textural features derived from image analysis techniques to select the best feature extraction approach.

- 2) Enhanced Generalizability with Feature Selection: To mitigate overfitting and improve model efficiency, HFF-Net employs a Mutual Information (MI) filter for feature selection.
- 3) PSO-Optimized SVM Classifier: The proposed HFF-Net approach employs a SVM classifier for the ultimate cell classification task. However, it incorporates PSO to fine-tune the SVM's hyperparameters, ensuring that the classifier is optimally configured to handle the specific feature space derived from the input data.

The article is organized as follows: Section II delves into related work, Section III details about the dataset used, Section IV discusses the methods deployed in the proposed work. To understand the effectiveness of these methods, Section V presents the experimental results along with a thorough discussion of their implications. Section VI concludes the overall contribution of the work.

II. RELATED WORKS

This section reviews existing research relevant to HFF-Net, focusing on studies that employ transfer learning techniques in conjunction with Convolutional Neural Networks (CNNs), as well as mixed techniques to improve the accuracy of cervical cancer detection through the analysis of Pap smear image data.

Zhao et al. [16] achieved impressive performance on the HERlev dataset with a deep CNN for end-to-end classification. Manna et al. [6] utilised CNN models such as Inception v3, Xception, DenseNet-169 for feature extraction and achieved high accuracy with a fuzzy rank-based fusion strategy. Basak et al. [17] proposed method for the selection of features extracted from CNN using principal component analysis method and a Grey Wolf Optimizer (GWO) for selecting informative features. Nguyen et al. [18] proposed a novel ensemble approach with pre-trained CNNs and averaging for potentially better accuracy on limited datasets. Maurya et al. [19] explored an ensemble approach combining CNNs and Vision Transformer (ViT) models, achieving an accuracy of 95.14%. These investigations demonstrate the promising capabilities of transfer learning methodologies and diverse approaches deploying convolutional neural networks for precisely classifying cervical cancer classes.

However, purely DL approaches can lose domain knowledge. To address this, hybrid approaches are emerging that combine DL with manually engineered features. Dong et al. [20] combined Inception v3 with handcrafted features to address limitations of traditional feature extraction and purely DL approaches for screening of cervical cancer classification. Attallah and Omneya [21] used multiple compact CNNs alongside features from various domains (spatial, time-frequency, statistical/textural). Jia et al. [22] leveraged a combination of informative GLCM and Gabor features and

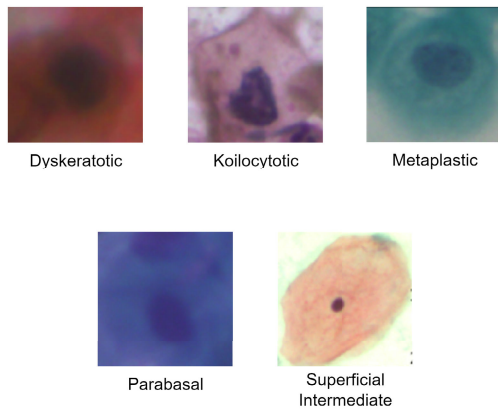


FIGURE 1. Sample image of each class in dataset.

a pre-trained LeNet-5 CNN for high-level abstractions. Their approach also incorporates dimensionality reduction and weighted fusion to optimize the features before feeding them into an SVM classifier for cell detection and classification. Natarajan et al. [23] explored pre-trained DCNNs (VGG16, ResNet50 etc.), analyzed the effect of classifiers (SVM, KNN etc.) on ConvNeXt-XL, and proposed a novel ConvNeXt-XL with custom designed and deep feature fusion for potentially better classification accuracy. Mathivanan et al. [24] utilized pre-trained deep learning models (AlexNet, ResNet-152, InceptionV3) combined with machine learning algorithms for cervical cancer detection. ResNet152 combined with Simple Logistic classifiers achieved 98.08% accuracy. Attallah and Omneya [25] combined the features obtained from the last three layer of the three different lightweight CNNs such as MobileNet, DarkNet-19, and ResNet-18 and the selected features were used to classify the Pap smear images of the cervical cancer using cubic SVM. In another work, Ali et al. [26] proposed an ensemble by combining the prediction made by four different machine learning classifiers such as Random Forest, Support Vector Machine, Naive Bayes, and Decision Tree, for cervical cancer classification. In other work Joynab et al. [27], proposed a learning-based convolutional neural network architectures which were used for the classification of Pap smear cervical cell images.

None of these methods utilised the domain specific manually engineered color, texture and shape features specifically related to the Pap smear cell images. Therefore, these domain-specific features termed as ‘artificial features’ through rest of the literature are combined with the more generic deep features extracted from the pre-trained deep CNN models.

III. DATASET USED

The suggested method is applied on the SIPaKMeD dataset [28], a publicly available collection of annotated cervical cytology images. The dataset encompasses 4,049 cervical cell images categorized into five distinct classes. An example image from each class is represented in Fig. 1. The breakdown of the class labels is provided in Table 1.

TABLE 1. Distribution of total images by classes.

Classes	Total Images
Dyskeratotic	813
Koilocytotic	825
Metaplastic	793
Parabasal	787
Superficial-intermediate	831
Total	4,049

TABLE 2. Distribution of images by classes.

Classes	Train	Valid	Total images
Dyskeratotic	634	179	813
Koilocytotic	674	151	825
Metaplastic	650	143	793
Parabasal	615	172	787
Superficial-intermediate	666	165	831
Total	3239	810	4049

A. DATASET SPLIT

The SIPaKMeD dataset was partitioned in an 80:20, training and validation split across all classes. as depicted in Table 2.

IV. PROPOSED METHODOLOGY

The proposed method has been termed ‘HFF-Net’ and the architecture of the proposed method has been shown in Fig. 2. This section elaborates how different domain specific features are extracted from an input cervical cell Pap smear images and further combined with the deep features. This section also explains the feature selection and classifier’s hyper parameter optimisation technique used in this work.

A. FEATURE EXTRACTION

The performance of the proposed method has been tested on deep features extracted from different pre-trained CNN models. However, the best performance has been obtained when the features extracted from the MobileNetV1 model were combined with the domain specific color, texture and shape features.

1) FEATURE EXTRACTION USING MOBILENETV1

MobileNetV1 [29] incorporates depthwise separable convolutions for efficient image classification on mobile and embedded devices. This factorization breaks down a standard convolution (applied with a single filter of size $H \times W \times C$, where C represents input channels, H denotes filter height and W is the width of the kernel) into two steps: depthwise and pointwise convolutions. In the depthwise stage, a lightweight filter of size $H \times W \times 1$ is applied to each input channel separately, significantly reducing computational cost compared to applying a single filter across all channels (C filters). This outputs C feature maps, one for each processed channel. The subsequent pointwise convolution employs a $1 \times 1 \times C$ filter to combine these C feature maps into a single output, effectively creating new features through linear combinations. This two-step approach achieves comparable feature extraction to standard convolutions while drastically reducing the number

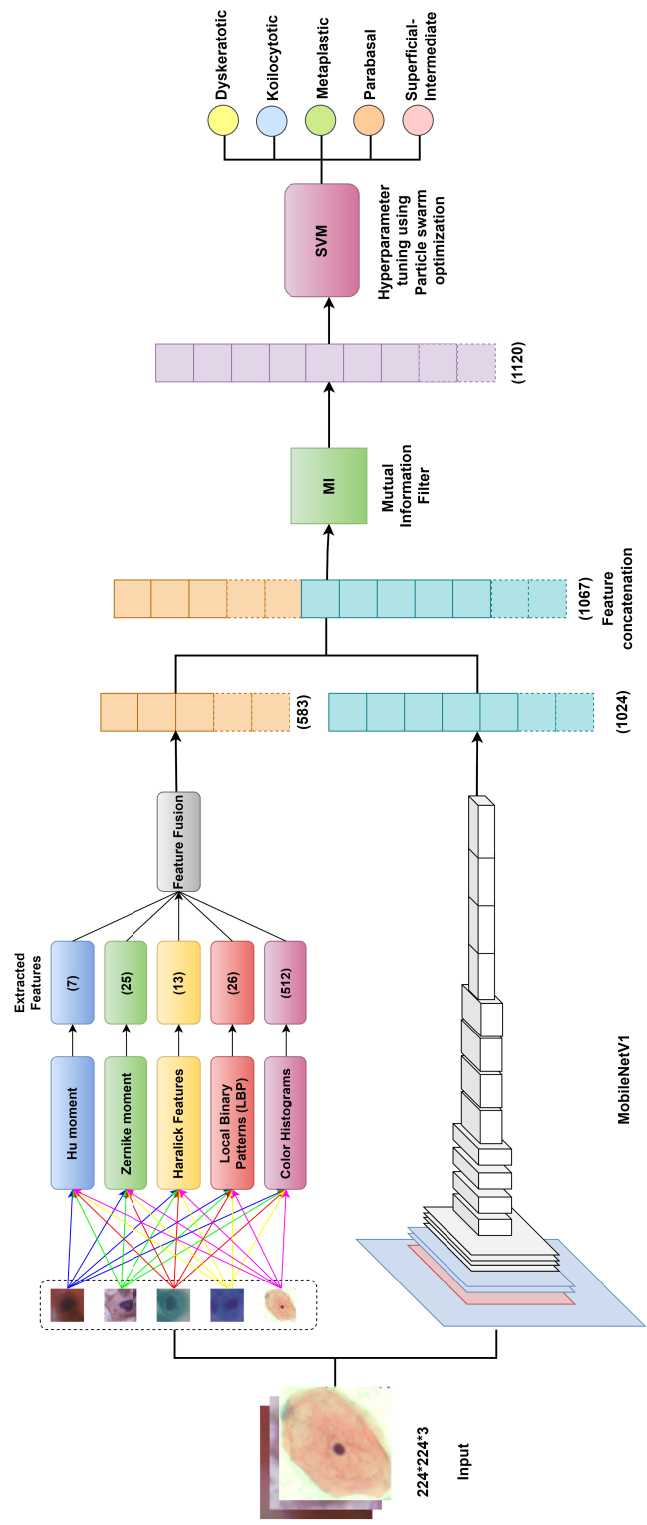


FIGURE 2. Detailed architecture of the proposed method.

of parameters and computations. MobileNetV1 stacks these depthwise separable convolution blocks along with global average pooling to reduce spatial dimensions and generate a compact feature vector for classification. This efficient

architecture allows MobileNetV1 to achieve high accuracy on various image recognition tasks while maintaining low latency and a small footprint, making it ideal for resource-constrained environments.

2) FEATURE EXTRACTION USING DOMAIN SPECIFIC FEATURES OF SHAPE, TEXTURE AND COLOR

Various domain specific color, texture and shape features were extracted from the cell images. These features as a whole are termed 'global features' through rest of the literature as these features explain the global properties of an input image when combined altogether. In this work, different shape features such as Hu and Zernike moments; texture features such as Harlick and Local Binary pattern features and color histogram features were studied to check their effectiveness in context to the present cervical cancer cell image classification task. A brief description of all these features have been explained as follows:

a: HU MOMENTS

Hu Moments [30] are a set of seven invariant moments derived from image moments that are used in image processing for shape analysis. Hu Moments helps in distinguishing cervical cancer cell images based on several geometric features present in them.

b: ZERNIKE MOMENTS

The Zernike moment [31] are a form of orthogonal invariant moments that is widely employed in extracting and describing image shape features. Due to their basis in orthogonal radial polynomials, these moments exhibit robustness against noise and minimize redundancy. Zernike moments of lower-order capture overall shape, while higher-order ones detail finer aspects, collectively providing a comprehensive representation of image shape features.

Let the image intensity function is denoted by $f(r, \theta)$. Zernike moment of order m with repetition n is defined as:

$$Z_{mn} = \frac{1}{\pi} \iint_{r \leq 1} f(r, \theta) \bar{V}_{mn}(r, \theta) r dr d\theta \quad (1)$$

Here, $\bar{V}_{mn}(r, \theta)$ is the complex conjugate of the Zernike polynomial $V_{mn}(r, \theta)$. Both m and n are integers satisfying the relation:

$$|m - n| \text{ is even and } |n| \leq |m| \quad (2)$$

and $V_{mn}(r, \theta)$ is expressed as:

$$V_{mn}(r, \theta) = e^{jm\theta} R_{mn}(r) \quad (3)$$

where $j = \sqrt{-1}$, and $R_{mn}(r)$ is the orthogonal radial polynomial which is given by:

$$R_{mn}(r) = \sum_{s=0}^{(m-|n|)/2} (-1)^s \frac{(m-s)!}{s! \left(\frac{m+|n|}{2} - s\right)! \left(\frac{m-|n|}{2} - s\right)!} r^{m-2s} \quad (4)$$

From the aforementioned formulas and Euler's complex number formula, the following equations can be derived:

$$R_{mn}(r) = R_{nm}(r) \quad (5)$$

$$V_{mn}(r, \theta) = V_{nm}(r, \theta) \quad (6)$$

Zernike moments offer a unique advantage in their ability to efficiently encode complex cell boundary information and texture details, which are vital cues for detecting malignant abnormalities in cervical cells.

c: HARALICK FEATURES

Texture is a fundamental visual property that plays a crucial role in image analysis and recognition. Haralick features provide a quantitative way to depict the spatial pixel intensities distribution within an snapshot, essentially capturing its textural characteristics [32]. These features are calculated based on a concept called a co-occurrence matrix (GLCM). The GLCM captures how often pairs of pixels with specific intensity values appear next to each other in a particular direction (e.g., horizontally, vertically, diagonally). By analyzing the GLCM, Haralick features extract various statistical properties that describe the texture of the image. These properties can include measures of contrast, homogeneity, entropy, and directionality. Haralick features are particularly beneficial for pap smear image classification of cervical cancer because the textural patterns of cell nuclei and cytoplasm can vary significantly between normal and abnormal cells, and these features can effectively capture these variations. Table 3 summarizes the haralick texture features.

TABLE 3. Haralick texture features.

Feature	Name of Texture
hf_1	Angular Second Moment (ASM)
hf_2	Contrast (CON)
hf_3	Correlation (COR)
hf_4	Sum of Squares: Variance (VAR)
hf_5	Inverse Difference Moment (IDM)
hf_6	Sum Average (SAV)
hf_7	Sum Variance (SVA)
hf_8	Sum Entropy (SEN)
hf_9	Entropy (ENT)
hf_{10}	Difference Variance (DVA)
hf_{11}	Difference Entropy (DEN)
hf_{12}, hf_{13}	Information Measures of Correlation (IMC)

d: LOCAL BINARY PATTERNS (LBP)

These patterns (LBP) [33] offer a powerful tool for capturing local texture information. LBP works by analyzing small neighborhoods around every pixel in the representation. It evaluates the intensity of the central pixel in relation to its neighboring pixels and represents this evaluation as a binary string. Pixels with higher intensity than the center contribute a 1 to the string, while those lower contribute a 0. This binary string is then transformed into a single decimal value, creating a label, effectively summarizing the local intensity pattern around the central pixel. The LBP operator is denoted as $LBP_{(J,K)}^u$, where J and K specify the size of the neighborhood (e.g., 3×3) and u defines that only uniform patterns where transitions from 0 to 1 (or vice versa) in the binary string occur at most twice. All remaining patterns are designated with a single label. Once the LBP labeling is applied to an image, resulting in a labeled image denoted as $g_l(s, t)$, we can

proceed to calculate the LBP histogram, denoted as H_i :

$$H_i = \sum_{(s,t)} I\{g_I(s,t) = i\}, \quad i = 0, 1, \dots, n-1 \quad (7)$$

where n is the number of LBP labels. It captures the spread of these LBP labels across the entire image. The LBP operator has computational simplicity; hence it is possible to analyze the images in real-time applications. LBP is effective for present work because its ability to encode local pixel intensity patterns makes it adept at capturing the intricate micro-textural details.

e: COLOR HISTOGRAMS

Color histograms [34] are a fundamental tool for characterizing the global distribution of pixel intensities within an image. It represents the statistical information of first-order of the image. The structure of the histogram provides a concise representation of an image's overall brightness, contrast, and tonal variations. Histograms also exhibit some degree of invariance to image translations and rotations. This means slight shifts or rotations in the image won't significantly alter the histogram. Additionally, normalization techniques can account for image scale variations, allowing for comparisons between images of different sizes. By analysing the intensity distribution, we can identify thresholds that effectively separate foreground objects from the background. In tasks like isolating cells of interest in pre-processed Pap smear images, this segmentation capability of histograms proves particularly useful.

B. FEATURE SELECTION USING MUTUAL INFORMATION (MI) FILTER

Feature selection assists in identifying the most significant features that may improve model performance in terms of computational speed, generalization, and interpretability. There are three major categories of feature selection methods [35]. A wrapper is a mechanism that involves the learning model in the task of feature selection, showing good generalization but being computationally expensive. Embedded methods integrate feature selection with the learning method, focusing on specific properties of models but requiring fewer computational resources. Deciding how many attributes or filters to use balances desired accuracy with computational resources; therefore, filter-based methods are generally more efficient.

Mutual Information (MI) [36] has emerged as a popular scoring function for filter-based feature selection methods. MI quantifies the statistical dependence between two variables- intuitively, it quantifies the extent to which understanding one variable decreases uncertainty about the other. MI captures nonlinear relationships between features and the target variable and is unaffected by invertible and differentiable transformations of the data, making it widely utilized for feature selection. It implements the relationship between a feature (denoted as M) and the target variable (denoted as N) to identify informative features. Intuitively,

MI measures the degree to which knowing the value of M diminishes uncertainty about N .

$$I(M, N) = \sum_{i,j} P(M = a_i, N = b_j) \log \frac{P(M = a_i, N = b_j)}{P(M = a_i)P(N = b_j)} \quad (8)$$

where $I(M, N)$ is the Mutual information score between M and N ; $P(M, N)$ is the joint probability distribution between M and N , and $P(M)$ and $P(N)$ are the probability distributions of feature M and target variable N , respectively. Features are then ranked based on their MI scores. Features with high MI scores are considered more relevant and informative, while those with low scores can be removed using a suitable threshold. This approach helps eliminate redundant or irrelevant features.

C. CATEGORIZATION USING PSO-TUNED SVM

SVMs [37] excel in multi-class classification tasks. The technique involves mapping data points to a higher-dimensional feature space, where they can be separated more effectively by a linear boundary. SVMs can effectively create a hyperplane (decision boundary) to categorize data points. This method aims to find the optimal hyperplane that maximizes the distance from the decision boundary to the closest data points (support vectors) while minimizing errors, which leads to models that generalize well to unseen data. The effectiveness and accuracy of an SVM is highly dependent on its hyperparameters, which control the learning process. Manually adjusting and optimizing these hyperparameters is often a tedious and labor-intensive process that requires significant effort. To address this challenge, we use PSO as an efficient method for hyperparameter tuning [38].

- 1) **Hyperparameter Tuning using PSO:** The PSO-Tuned SVM [39] algorithm integrates Particle Swarm Optimization (PSO) with Support Vector Machines (SVM) to optimize hyperparameters critical for categorization performance. The SVM model utilizes an RBF kernel, with two key hyperparameters: the penalty parameter C and the kernel coefficient γ . PSO initializes a swarm of particles representing potential hyperparameter combinations within predefined bounds and evaluates their fitness through the accuracy of the SVM. The position and velocity of each particle are updated iteratively based on individual and global best positions, boosting exploration and convergence towards optimal solutions. The process continues until a maximum number of iterations is reached or convergence occurs, resulting in optimal values for C and γ . The proposed PSO-tuned SVM method has been presented as Algorithm 1.

V. EXPERIMENTAL RESULTS AND DISCUSSIONS

The experimentation details and corresponding results are discussed in this section:

Algorithm 1 PSO-Tuned SVM**Input:** Number of particles N_p , Maximum iterations T_{max} **Initialize:**

Define bounds:

$$lb = \begin{bmatrix} 1 \\ 0.0001 \end{bmatrix}, \quad ub = \begin{bmatrix} 100 \\ 1 \end{bmatrix}$$

Set $C \in [1, 100]$ and $\gamma \in [0.0001, 1]$ **Load dataset:** Feature matrix $X \in \mathbb{R}^{m \times n}$, label vector $y \in \mathbb{R}^m$ **Define objective function:**

$$f(\text{params}) = -\frac{1}{5} \sum_{k=1}^5 \text{Accuracy}_k$$

where $\text{params} = [C, \gamma]$ for SVM:SVC($\text{kernel} = \text{'rbf'}$, $C = \text{params}[0]$, $\gamma = \text{params}[1]$)**Initialize particles:****for** $i = 1$ to N_p **do**

Randomly initialize:

$$\text{pos}[i] \sim U(lb, ub), \quad \text{vel}[i] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Set $pbest[i] = \text{pos}[i]$ **end for**Initialize $gbest$ **Optimization loop:****for** $t = 1$ to T_{max} **do****for** $i = 1$ to N_p **do**

Update velocity:

$$\text{vel}[i] = w \cdot \text{vel}[i] + c_1 \cdot r_1 \cdot (pbest[i] - \text{pos}[i]) + c_2 \cdot r_2 \cdot (gbest - \text{pos}[i])$$

Update position:

$$\text{pos}[i] = \text{Clamp}(\text{pos}[i] + \text{vel}[i], lb, ub)$$

Evaluate score = $f(\text{pos}[i])$ **if** score < $f(pbest[i])$ **then** $pbest[i] = \text{pos}[i]$ **end if****if** score < $f(gbest)$ **then** $gbest = \text{pos}[i]$ **end if****end for****end for****Output:** Optimal hyperparameters $gbest$ and score $f(gbest)$ **A. EXPERIMENTAL SETUP**

The experiments for this work were conducted on a computing system facilitated with an Nvidia P100 graphics processing unit (GPU) and 12GB of RAM. The proposed methodology was implemented using the Keras 2.14.0 library within the Python programming environment.

B. IMPLEMENTATION

- 1) **SVM Hyperparameter Tuning:** Classification was done using SVM classifier. To optimize the hyperparameters of the SVM classifier, PSO is utilized to finalise the optimal values for the kernel function, regularization parameter and kernel coefficient as shown in Table 4.

TABLE 4. Hyperparameters of the proposed architecture.

Hyperparameters	Values
Kernel	rbf
Regularization parameter (C)	10
Kernel coefficient γ	0.001

- 2) **Performance analysis of pretrained CNNs for optimal feature extraction:** To identify the most efficient CNN model for feature extraction evaluation was done on a broad spectrum of pre-trained convolutional neural networks (CNNs), including established architectures like VGG19, ResNet variants, and DenseNets as shown in Table 5. While several models achieved promising accuracy exceeding 90%, MobileNetV1 emerged as the leader with the highest accuracy of 95% using features extracted from its last convolutional layer (Conv_pw_13). The better performance of MobileNetV1 stems from its optimized architecture, employing depth-wise separable convolutions to balance computational efficiency with robust feature representation. This structure extracts both low-level morphological and high-level abstract features, crucial for differentiating subtle cytological variations in cervical cell types.

- 3) **Evaluating the Performance of Combined Shape, Texture, and Color Features for Feature Extraction:** Then, to complement the DL features extracted from MobileNetV1, manually engineered were studied, including those based on image moments (Hu and Zernike moments), texture analysis (Haralick features and Local Binary Patterns - LBP), and global color distribution (Color Histogram) as described in Table 6. While individual features provided moderate accuracy (around 43% - 71%), the most promising results were achieved by combining these features. Notably, the combination of all the features (#1 - #5) achieved an accuracy of 95%, matching the performance of MobileNetV1 alone.

Further, various combinations of MobileNetV1's output features and tailored image features were investigated as shown in Table 7. These combinations resulted in promising accuracy. The best performing combination along with all engineered features (MobileNetV1(Conv_pw_13)+Global features) achieved the highest accuracy of 97.16%. This significant improvement highlights the effectiveness of combining abstract features extracted using DL

TABLE 5. Performance Comparison of Deep CNN Architectures with last convolutional features and with last fully connected layer features.

S.No.	Deep CNN Model	Accuracy with last conv. layer features	Accuracy with last fully connected layer features
1	DarkNet53	94.82	94.34
2	SEResNet152	94.28	94.14
3	SEResNet10	92.46	91.29
4	SEResNet18	92.33	90.67
5	SEResNet26	91.59	91.20
6	Res34	91.43	90.58
7	SE154	92.12	90.76
8	SE28	90.85	91.47
9	VGG19	91.96	94.38
10	VGG16	91.74	93.21
11	VGG13	91.68	92.35
12	ResNet152	94.13	91.49
13	ResNet201	92.25	90.98
14	SEResNet101	94.72	93.66
15	DenseNet161	94.58	93.40
16	DenseNet201	93.37	91.23
17	InceptionResNetV2	90.89	90.22
18	AlexNet	92.13	93.45
19	MobileNetV2	92.67	91.78
20	MobileNetV1 (Conv_pw_13)	95.00	94.87

TABLE 6. Performance of manually designed features and their combinations.

S.No.	Manually designed features	Type of descriptor	#Features	Accuracy
1	Hu moment	S	7	43
2	Zernike moment (radius=21, degree=8)	S	25	43
3	Haralick features	T	13	68
4	Lbp (numPoints=24, radius=8)	T	26	71
5	Color Histogram (bins=8)	C	512	91
6	Global feature (#1,#2,#3,#4,#5,)	S,S, T,T, C	583	95
7	#4,#5	T,C	538	94
8	#3,#4,#5	T,T,C	551	75
9	#1,#2	S,S	32	53
10	#1,#2,#3	S,S,T	45	69
11	#1,#2,#3,#4	S,S,T,T	71	74

methods and strong feature extraction using image descriptors for feature extraction.

The PSO-tuned SVM outperforms standalone fine-tuned CNN models when trained with a combination of deep features and domain-specific handcrafted features due to the complementary strengths of these approaches. Deep features extracted from CNNs capture complex, hierarchical patterns in the data, while handcrafted features capturing color, texture and shape related features present in the cytology images of the cervical cancer; thereby, provide valuable domain-specific insights. This integration allows the model to utilize multi-faceted information for improved categorization of cervical cancer cell images. Additionally, the PSO optimization ensures finely tuned hyperparameters for the SVM, enhancing its ability to generalize well on the unseen test set. These components when combined together provides a more robust feature representation, leading to superior classification performance.

- 4) **Analysis of impact of introducing MI filter:** The impact of feature selection on the performance of model is discussed in this section. By setting the parameter K to 1100, the MI filter selected the most

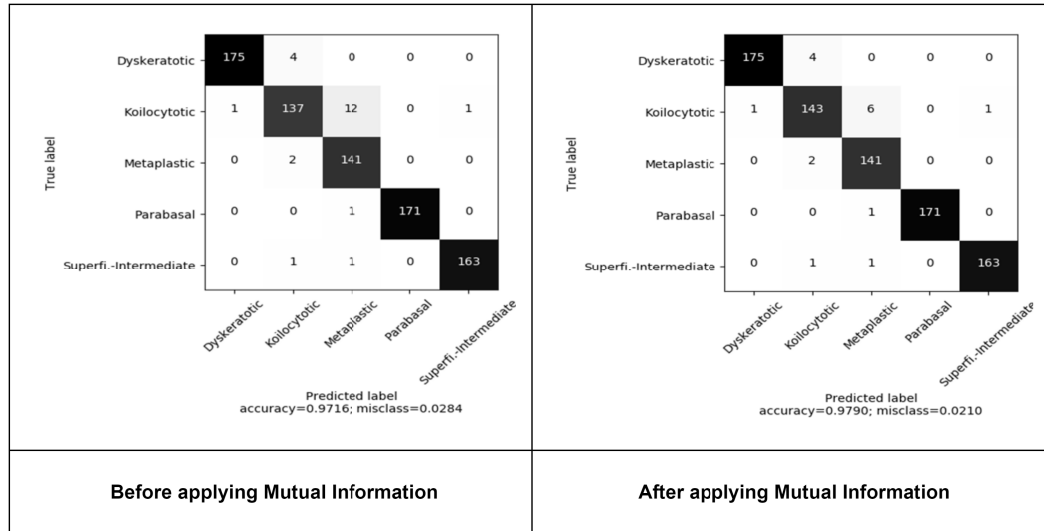
informative features. A comparison of the confusion matrices before and after applying the MI filter shows a favourable outcome on classification performance as shown in Figure 3. The overall improvement in accuracy from 97.16% to 97.9% suggests the effectiveness of feature selection.

Tables 8 and 9 present the classification metrics (precision, recall, F1-score, and support) for all five classes, before and after applying the MI filter. These metrics provide valuable observations into the model's performance for individual cell type. As observed in the tables, the model achieved high precision and recall values for all classes, indicating its ability to accurately classify cervical cells.

For a more comprehensive evaluation, ROC curves are displayed in Fig. 4. These curves illustrate the trade-off between true positive rate (TPR) and false positive rate (FPR) across different classification thresholds. A larger Area Under the Curve (AUC) indicates superior model performance. Analysing the ROC curves before and after applying the MI filter provides further insights into the model's ability to discriminate between different cell types.

TABLE 7. Performance of fused features.

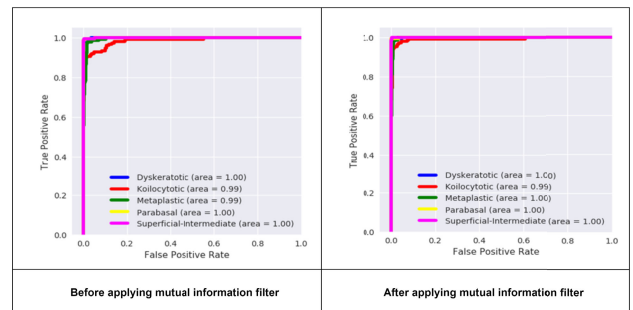
Fused Features	Accuracy
MobliNetV1(Conv_pw_13), LBP Texture Features	95%
MobliNetV1(Conv_pw_13), Haralick Texture Features	96%
MobliNetV1(Conv_pw_13), Hu Moment Shape + LBP Texture	96.2%
MobliNetV1(Conv_pw_13), Haralick Texture + Zernike Moment Shape	96.7%
MobliNetV1(Conv_pw_13) + Global Features (Hu Moment, Zernike Moment, Haralick, LBP, Color Histogram)	97.16%

**FIGURE 3.** Confusion Matrices before and after applying MI filter.**TABLE 8.** Performance metrics before applying MI filter.

Class	Precision	Recall	F1-Score	Support
Dyskeratotic	0.99	0.98	0.98	179
Koilocytotic	0.94	0.91	0.93	151
Metaplastic	0.91	0.98	0.94	143
Parabasal	1.00	0.99	0.99	172
Superficial-intermediate	0.99	0.99	0.99	165

TABLE 9. Performance metrics after applying MI filter.

Class	Precision	Recall	F1-Score	Support
Dyskeratotic	0.99	0.98	0.99	179
Koilocytotic	0.95	0.95	0.95	151
Metaplastic	0.95	0.99	0.97	143
Parabasal	1.00	0.99	1.00	172
Superficial-intermediate	0.99	0.99	0.99	165

**FIGURE 4.** ROC curves before and after applying MI filter.**TABLE 10.** Comparison of training time before and after applying MI filter.

Time (seconds)	Before applying MI filter	After applying MI filter
Training Time	703.45	471
Testing Time	1.69	1.35

The application of the MI filter significantly reduced the training time. Table 10 summarizes the training and testing times before and after applying the MI filter. As evident, the training time decreased from 703.45 seconds to 471 seconds after applying the MI filter. This reduction can be attributed to the smaller feature set used by the model, leading to faster training. The testing time also experienced a

slight improvement, reducing from 1.69 seconds to 1.35 seconds.

The feature selection process using MI effectively reduces noise in the data, which can otherwise lead to overfitting. By focusing on the most informative features, the model

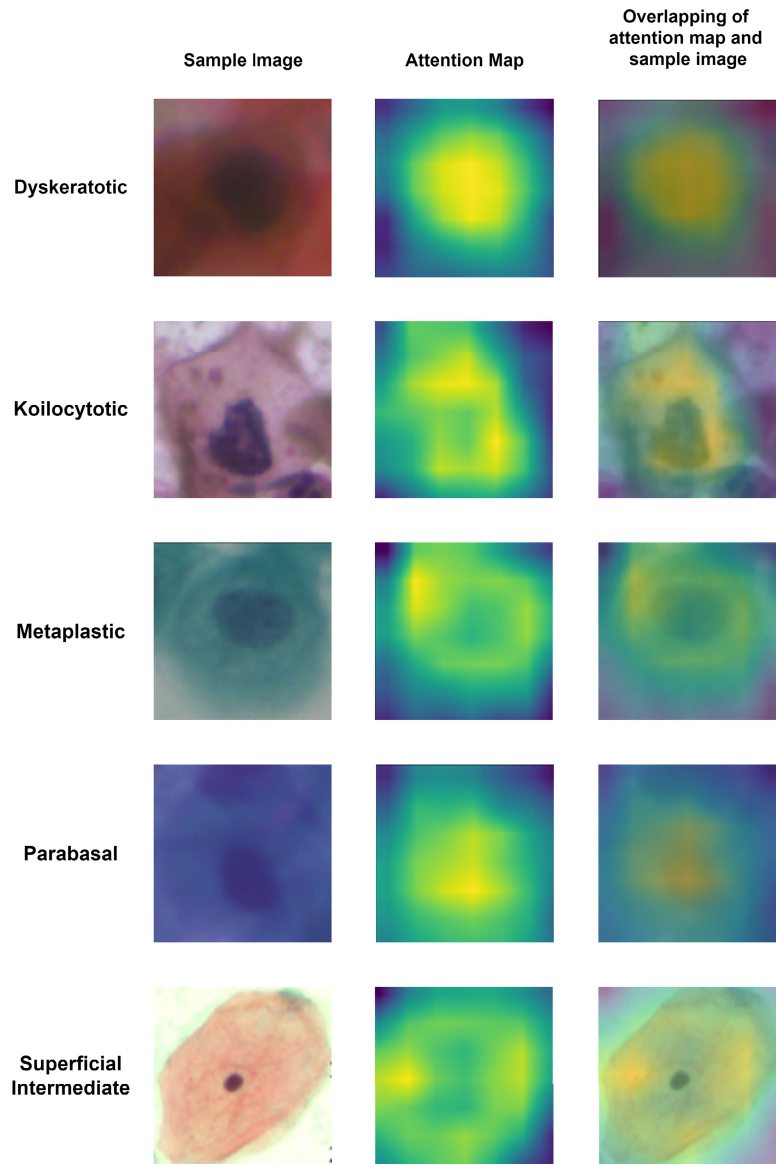


FIGURE 5. A visual representation of heat maps generated from the last convolutional layer of MobileNetV1, highlighting regions of interest and their overlap on the original images.

becomes more generalizable and less likely to be influenced by false patterns in the training data.

C. VISUALIZATION

The t-SNE plots, a dimensionality reduction technique, provide a visual representation of the data distribution before and after applying the MI filter. The t-SNE plot, after applying the MI filter demonstrated tighter and more well-defined clusters for each cell type as depicted by Fig. 6.

This confirms the effectiveness of the MI filter in reducing dimensionality by selecting the most informative features. These features capture the essential characteristics that differentiate between different cell types and enhanced classification performance by focusing the model on the most relevant features.

In Fig. 5, visualization using heat maps is shown to directly highlight critical regions within images. By superimposing these heat maps onto the original images, the highlighted features aids in understanding patterns, trends, and focal points crucial for analysis and interpretation.

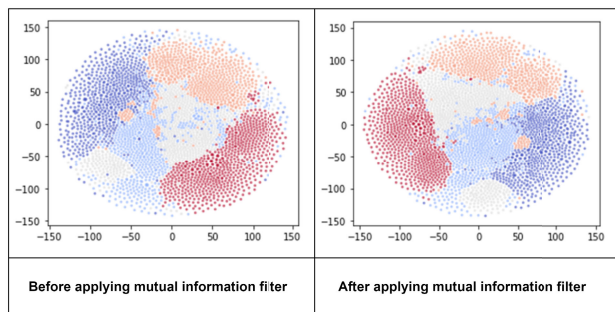
D. COMPARATIVE ANALYSIS

The proficiency of the suggested method for cervical cell classification is evaluated and compared with existing approaches. This comparison is crucial to evaluate the effectiveness of HFF-Net and its contribution to the field. Table 11 summarizes the performance (accuracy) of various approaches reported in recent literature.

Building on extensive experimentation to optimize component performance and efficiency, proposed architecture,

TABLE 11. Comparison of accuracy among different methods.

Author	Methodology Adopted	Accuracy (%)
O. Attallah [21]	ResNet-18 + DarkNet-19 + MobileNet + SVM	97.70
Maurya et al. [19]	ViT + CNN	97.65
	CNN + LSTM	95.80
Premanik et al. [40]	MobileNet + Inception + InceptionResNet	96.47
Yaman and Tuncer [41]	DarkNet-19 + NCA + SVM	96.70
Sahoo et al. [42]	Pre-trained models + Fuzzy rank-based ensemble + Advanced augmentation techniques	97.18
Basak et al. [17]	ResNet-50 + DenseNet-121 + Inception + VGG-16 + PCA + GWO	97.80
Proposed Method HFF-Net	MobileNetV1 + Manually designed features, MI filter, SVM (PSO)	97.90

**FIGURE 6.** t-SNE plot visualization of cluster of classes before and after MI filter.

HFF-Net obtains significant performance while offering several advantages such as light weight design, low latency and enhanced interpretability.

VI. CONCLUSION

The work proposes a novel approach for cervical cell classification that integrates the strengths of both DL and domain-specific features. MobileNetV1, a CNN architecture known for its efficiency, extracts deep features from Pap smear images. These features are complemented by features derived from image moments, texture analysis, and color distribution. Feature selection using a Mutual Information (MI) filter optimizes the model by reducing dimensionality and focusing on the most informative features. The combined features are then fed into a SVM classifier optimized using PSO for final cell classification. This approach achieves a competitive accuracy of 97.9%, comparable to several top performing methods. The combination of efficient DL features, informative domain specific features, and dimensionality reduction using MI filtering demonstrates a promising strategy for accurate and efficient cervical cell classification, potentially improving early detection rates and patient outcomes.

DECLARATION OF COMPETING INTEREST

The authors affirm that they do not have any competing financial interests or personal relationships that could have influenced the reported work in this paper.

DATA AVAILABILITY

The authors have also stated that data will be accessible upon request.

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